

# Financial Econometrics

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Introduction  
Statistical properties of financial returns

**Financial Econometrics** : a set of statistical methods aiming at analyzing and testing the relations/models provided by the Financial theory.

Financial econometrics is a relatively new topic (beginning of the 1990s).

An academic/practitioner's topic which is visible through

- textbooks
- an academic society : SoFiE
- conferences

In most chapters of this course : static models (no serial dependencies).

Dynamic models will be studied in the last chapter and in other courses.

## Chapter I : Statistical properties of financial returns

- ① Returns of a financial asset
- ② Statistical properties
  - ① Sample paths of prices and returns
  - ② Marginal distributions of returns
  - ③ Serial dependencies and assumptions on the returns processes

## Chapter II : Efficient portfolios

- ① Efficient frontier
- ② Sharpe performance
- ③ Estimation
  - ① Asymptotic properties of empirical moments
  - ② Estimation of the Sharpe performance
  - ③ Estimation of the efficient frontier
- ④ Tests
  - ① Test of the equality of performance of two portfolios
  - ② Efficiency test of a portfolio

## Chapter III : CAPM

- ① Regression of the baseline returns
- ② Systemic risk, diversification
- ③ CAPM
  - ① Equilibrium model
  - ② Regression on the market portfolio
- ④ Estimation
- ⑤ Tests of the CAPM
- ⑥ Application

## Chapter IV : Factor models

- ① Model and absence of arbitrage opportunities
- ② Estimation
- ③ Factors selection
  - ① Principal Component Analysis ; Factor analysis
  - ② Application

## Chapter V : Dynamic factor models

- ① Motivations
- ② Dynamic conditional betas
- ③ Dynamic Cholesky model

## Textbooks

-  Campbell, J.Y., Lo, A. and A.C. MacKinlay  
*The econometrics of financial markets.*  
Princeton University Press, 1997.
-  Gouriéroux, C., O. Scaillet and A. Szafarz  
*Econométrie de la finance.*  
Economica, 1997.
-  Gouriéroux, C., and J. Jasiak  
*Financial Econometrics : problems, models and methods.*  
Princeton University Press, 2001.
-  Linton, O.  
*Financial Econometrics : models and methods.*  
Cambridge, 2019.

- 1 Returns of a financial asset
- 2 Statistical properties



## Returns of a financial asset

Different types of financial assets : stocks, bonds, currencies, commodities, forward contracts...

$p_{i,t}$  : price of asset  $i$  at time  $t$

Net return from time  $t$  to  $t+1$  :

$$r_{i,t+1} = \frac{p_{i,t+1} - p_{i,t}}{p_{i,t}}.$$

Gross return from time  $t$  to  $t+1$  :

$$y_{i,t+1} = \frac{p_{i,t+1}}{p_{i,t}} = r_{i,t+1} + 1.$$

## Returns of a financial asset

Compound return (or log return) from time  $t$  to  $t+1$  :

$$\epsilon_{i,t+1} = \log \frac{p_{i,t+1}}{p_{i,t}} = \log(1 + r_{i,t+1}) \approx r_{i,t+1}.$$

Risk-free rate from time  $t$  to  $t+1$  :  $r_t$ .

Excess return of asset  $i$  from time  $t$  to  $t+1$  :

$$\tilde{r}_{i,t+1} = r_{i,t+1} - r_t.$$

**Remark** : those definitions have to be modified to account for dividends (for stocks) or coupons (for bonds). For instance :

$$r_{i,t+1} = \frac{p_{i,t+1} + d_{i,t+1} - p_{i,t}}{p_{i,t}}.$$

However, dividends are paid once a year and are difficult to include in the data. For yearly data, they have to be taken into account.

## Temporal aggregation on $k$ periods

Holding period for an investment is often more than 1 time unit.

Gross return from time  $t$  to  $t+k$  :

$$\begin{aligned} y_{i,t+k}^{(k)} &= \frac{p_{i,t+k}}{p_{i,t}} := r_{i,t+k}^{(k)} + 1 \\ &= \prod_{j=1}^k (r_{i,t+j}^{(1)} + 1). \end{aligned}$$

Annualized return : the implied one period constant net return

$$\bar{r}_{i,t+k}^{(k)} = \left\{ \prod_{j=1}^k (r_{i,t+j}^{(1)} + 1) \right\}^{1/k} - 1.$$

Compound return :

$$e_{i,t+k}^{(k)} = \log \frac{p_{i,t+k}}{p_{i,t}} = \sum_{j=1}^k e_{i,t+j}^{(1)}.$$

## Aggregation of assets

Let a portfolio with  $N$  assets : for  $i = 1, \dots, N$ , let  $a_i$  the number of units of asset  $i$  (assumed to be constant between  $t$  and  $t+1$ )

Value of the portfolio at time  $t$  :

$$V_t(\mathbf{a}) = \sum_{i=1}^N a_i p_{i,t} = \mathbf{a}' \mathbf{p}_t, \quad \mathbf{a} = (a_1, \dots, a_N)', \quad \mathbf{p}_t = (p_{1,t}, \dots, p_{N,t})'.$$

Value of the portfolio at time  $t+1$  :

$$V_{t+1}(\mathbf{a}) = \sum_{i=1}^N a_i p_{i,t+1}.$$

Net return of the portfolio :

$$r_{t+1}(\mathbf{a}) = \frac{V_{t+1}(\mathbf{a})}{V_t(\mathbf{a})} - 1 = \sum_{i=1}^N a_{i,t}^* r_{i,t+1}, \quad a_{i,t}^* = \frac{a_i p_{i,t}}{\sum_{j=1}^N a_j p_{j,t}}.$$

## Aggregation of assets

The compound return of a portfolio does not write as a linear combination of individual returns.

However, we have

$$\begin{aligned}\epsilon_{t+1}(\mathbf{a}) &:= \log \frac{V_{t+1}(\mathbf{a})}{V_t(\mathbf{a})} = \log \left\{ \sum_{i=1}^N a_{i,t}^* (r_{i,t+1} + 1) \right\} \\ &= \log \left\{ \sum_{i=1}^N a_{i,t}^* e^{\epsilon_{i,t+1}} \right\}.\end{aligned}$$

## 1 Returns of a financial asset

## 2 **Statistical properties**

- Characteristics of the returns distributions
- Stable distributions
- Dependencies

# "The behavior of stock market prices", J. of Business

Eugene Fama, 1965

Fama [Nobel prize in (2013)] promoted the theory that financial prices follow a random walk. In other words, price variations are independent and identically distributed (i.i.d.).

Hence, future prices are not more predictable than the sum of random numbers.

An answer to the chartist view which postulated that past prices are informative about their future values.

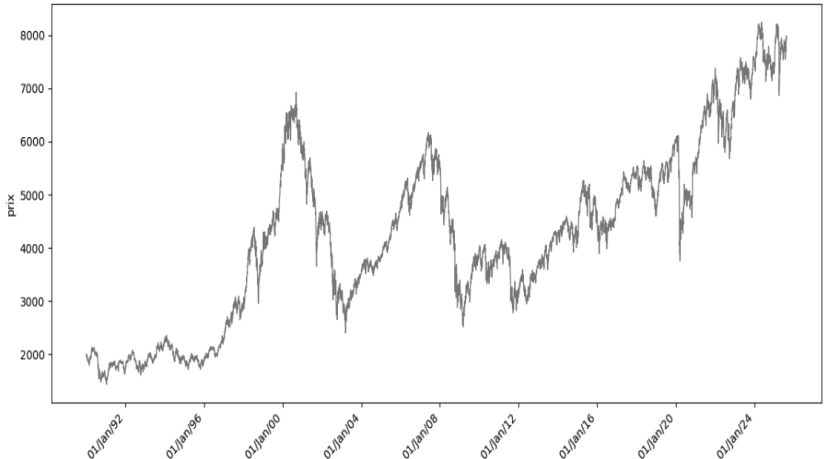
## Sample paths

- **Prices** : sample paths close to that of a random walk without intercept.
- **Returns/log-returns** : sample paths are much less erratic.  
Generally compatible with stationarity.

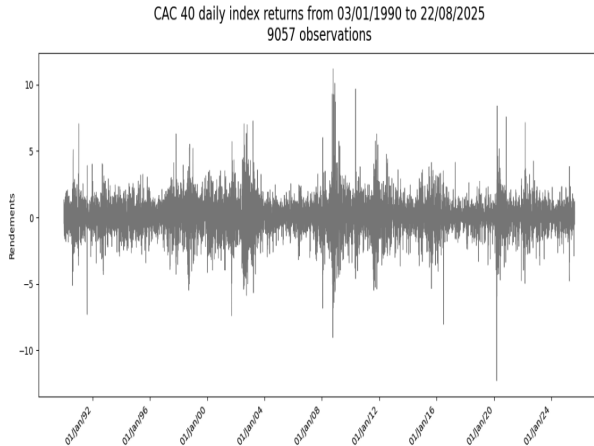


# CAC 40 daily index returns

CAC 40 daily index from 02/01/1990 to 22/08/2025  
9058 observations

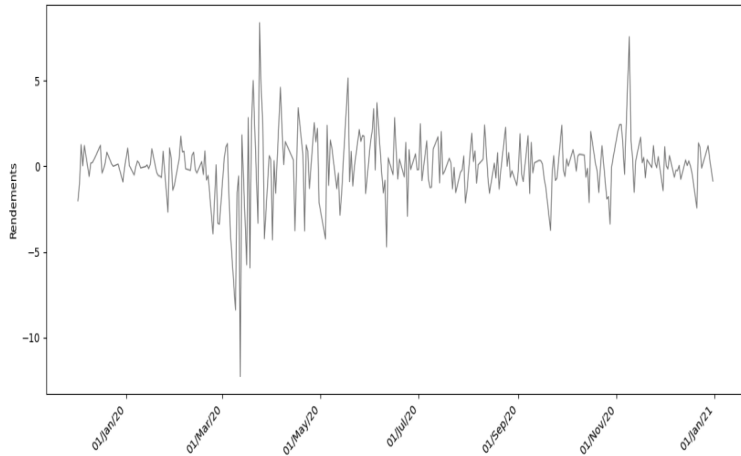


# CAC 40 daily index returns from 01/1990 to 08/2025



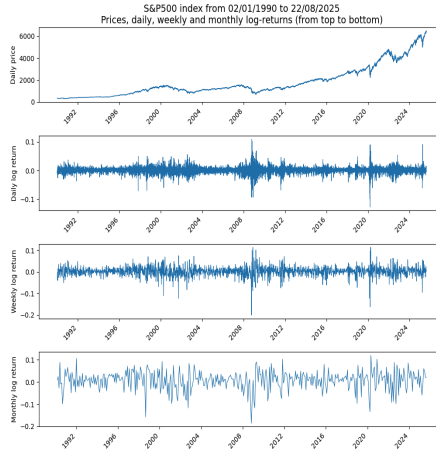
## CAC 40 index from 12/2019 to 12/2020

CAC 40 daily index returns from 01/12/2019 to 31/12/2020  
277 observations



# S&P500 index from 01/1990 to 08/2025

Prices, daily, weekly and monthly log-returns (from top to bottom)



## Marginal distributions

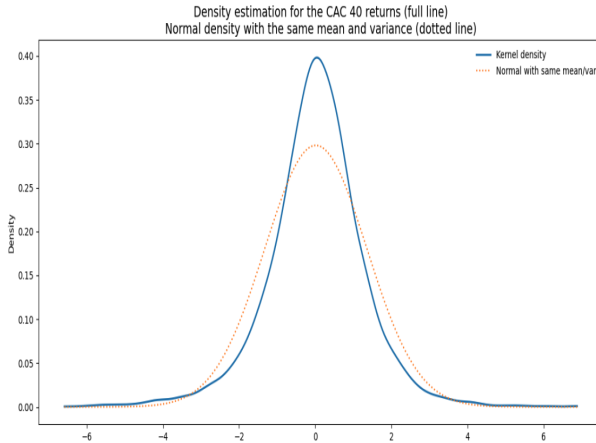
- Generally very different from a Gaussian.
- Leptokurtic distributions, peaked at zero and with heavy tails.
- An estimator of the marginal density is

$$\hat{f}_n(z) = \frac{1}{nh_n} \sum_{t=1}^n K\left(\frac{x_t - z}{h_n}\right)$$

where  $K$  is a density on  $\mathbb{R}$  and  $h_n$  is a smoothing parameter called the bandwidth.

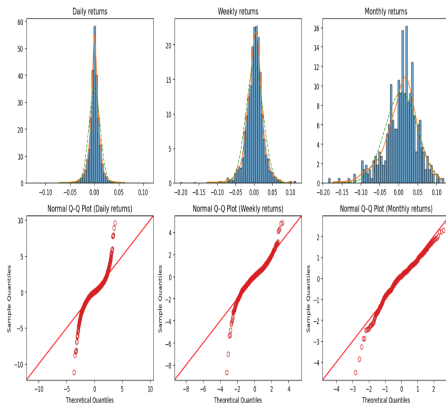
# Density estimation for the CAC 40 returns (full line)

Normal density (dotted line)



# Histograms and Q-Q plots of the S&P500 index returns 01/1990 to 08/2025

Histograms and Q-Q plots of the S&P500 Index returns  
02/01/1990 to 22/08/2025



## Some basics on stable distributions

- A probability measure  $\mu$  on  $\mathbb{R}$  is said to be **stable** if for every  $a > 0$  and  $b > 0$ , there exists  $c > 0$  and a real number  $e$  such that

$$aX_1 + bX_2 \stackrel{d}{=} cX + e,$$

where  $\mathcal{L}(X_1) = \mathcal{L}(X_2) = \mathcal{L}(X) = \mu$ , and  $X_1$  and  $X_2$  are independent.

- Equivalently :  $\mu$  is **stable** iff for each  $k \in \mathbb{N}$ , there are  $b_k > 0$  and  $e_k$  such that

$$X_1 + \dots + X_k \stackrel{d}{=} b_k X + e_k,$$

where  $\mathcal{L}(X_1) = \dots = \mathcal{L}(X_k) = \mathcal{L}(X) = \mu$ , and  $X_1, \dots, X_k$  are independent.

Moreover, there exists  $\alpha \in [0, 2]$  such that  $b_k = k^{1/\alpha}$  and we say that  $\mu$  is  **$\alpha$ -stable**.



## Some basics on stable distributions

Univariate  $\alpha$ -stable distributions are characterized by four parameters :

- $\alpha \in [0, 2]$  is called the **index of stability**, or tail exponent ;
- an asymmetry parameter  $\beta \in [-1, 1]$ ,
- a scale parameter  $\sigma \in (0, \infty)$ ,
- a location parameter  $m \in \mathbb{R}$ .

Thus we can write  $X \sim \mathcal{S}(\alpha, \beta, \sigma, m)$  for a variable following a stable distribution. If  $m = 0$  the distribution is called **strictly stable**.

The location and scale parameters are such that, if  $X \sim \mathcal{S}(\alpha, \beta, 1, 0)$ , then  $Y = \sigma X + m \sim \mathcal{S}(\alpha, \beta, \sigma, m)$ , where  $\sigma > 0$ .

## Some basics on stable distributions

In general, the probability density function (pdf) of a stable distribution is not known explicitly, but its characteristic function  $\psi(s) = E(e^{isX})$  has the closed form :

$$\log \psi(s) = -\sigma^\alpha |s|^\alpha \left\{ 1 - i\beta (\text{sign } s) \tan\left(\frac{\pi\alpha}{2}\right) \right\} + i\mu s,$$

if  $\alpha \neq 1$ , and

$$\log \psi(s) = -\sigma |s| \left\{ 1 + i\beta (\text{sign } s) \frac{2}{\pi} \log |s| \right\} + i\mu s,$$

if  $\alpha = 1$ .

## Some basics on stable distributions

There exist other parameterizations for  $\psi$ , but this one presents the advantage that

$$f_{\theta}(x) := (2\pi)^{-1} \int_{\mathbb{R}} \exp\{-is(x - \mu)\} \psi_{\alpha,\beta}(\sigma s) ds$$

is differentiable with respect to both  $x \in \mathbb{R}$  and

$\theta \in \Lambda := (0, 2) \times (-1, 1) \times (0, \infty) \times \mathbb{R}$  (see Nolan, 2003). Let  $f_{\alpha,\beta}$  be the **stable density of parameter  $\theta = (\alpha, \beta, 1, 0)$** . We have for  $\alpha \neq 1$ ,

$$f_{\alpha,\beta}(x) = \frac{1}{\pi} \int_0^{\infty} e^{-s^{\alpha}} \cos\left\{sx + \beta \tan\left(\frac{\pi\alpha}{2}\right)(s - s^{\alpha})\right\} ds$$

and for  $\alpha = 1$ ,

$$f_{\alpha,\beta}(x) = \frac{1}{\pi} \int_0^{\infty} e^{-s} \cos\left(sx + s\beta \frac{2}{\pi} \log s\right) ds$$

From the expansion  $(1 - s^{\alpha-1}) \tan\left(\frac{\pi\alpha}{2}\right) = \frac{2}{\pi} \log s + o(\alpha - 1)$ , the continuity at  $\alpha = 1$  is clear.

## Some basics on stable distributions

Note that  $f_{\theta}(x) = \sigma^{-1} f_{\alpha, \beta} \{ \sigma^{-1}(x - \mu) \}$  can be numerically evaluated from the previous expressions, or alternatively using the function `dstable()` of the R package `fBasics`.

### Particular cases :

- **Gaussian distribution**  $N(\mu, \sigma^2)$  : obtained for  $\alpha = 2$  ( $\beta$  non identifiable).
- **Cauchy distribution**  $\mathcal{C}(\mu, \sigma)$  : obtained for  $\alpha = 1$  and  $\beta = 0$ .
- **Lévy distribution** : obtained for  $\alpha = 1/2$  and  $\beta = 1$ . Density (when  $\sigma = 1$  and  $\mu = 0$ ) :

$$f(x) = \frac{1}{\sqrt{2\pi}} \frac{1}{x^{3/2}} \exp\left(\frac{-1}{2x}\right) \mathbf{1}_{x>0}.$$

## Some basics on stable distributions

The coefficient  $\alpha$  determines the tail of the distribution of  $X \sim S(\alpha, \beta, \sigma, \mu)$  in the sense that, when  $\alpha < 2$ , as  $x \rightarrow \infty$ ,

$$P(X < -x) \sim c_\alpha(1 - \beta)x^{-\alpha} \quad \text{and} \quad P(X > x) \sim c_\alpha(1 + \beta)x^{-\alpha},$$

with  $c_\alpha > 0$ .

Moreover, when  $X \sim S(\alpha, \beta, \sigma, \mu)$  with  $\alpha < 2$ ,

$$E|X|^p < \infty \quad \text{if and only if} \quad p < \alpha$$

For further details on stable variables see

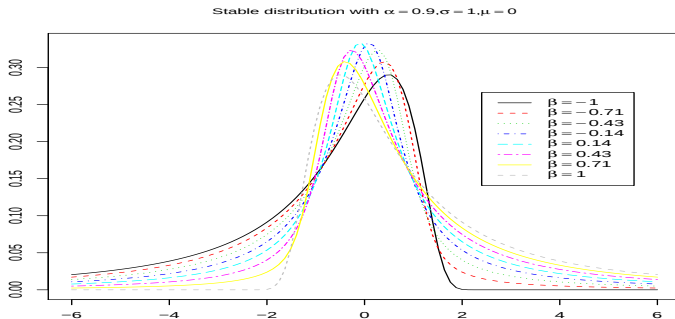


Samorodnitsky and Taqqu

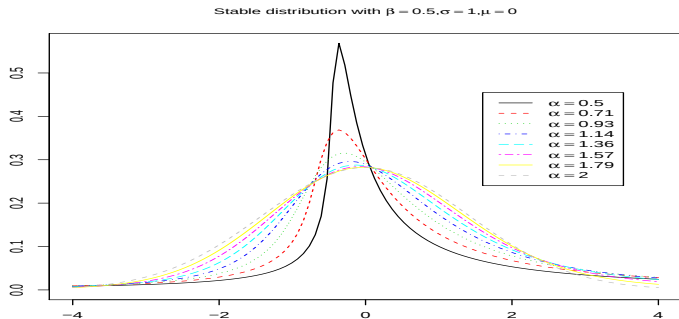
*Stable Non-Gaussian Random Processes.*

Chapman & Hall, 1994.

## Stable densities with different symmetry parameters



# Stable densities with different tail index parameters



$$E|X|^r < \infty \text{ iff } r < \alpha \text{ (when } \alpha < 2)$$

# Sums of i.i.d. converge to stable laws

## Central limit theorem

- Let  $(X_t)$  be an i.i.d. sequence.
- If there exists constants  $B_n > 0$  and  $A_n \in \mathbb{R}$  such that

$$\frac{X_1 + \dots + X_n - A_n}{B_n} \xrightarrow{d} \mu$$

then  **$\mu$  is stable** (strictly stable if  $A_n = 0$ ).

- The sequence  $(B_n)$  is  **$1/\alpha$ -regularly varying** :

$$B_n = n^{1/\alpha} L(n), \quad 0 < \alpha \leq 2$$

where  $L$  is a slowly varying function.

- The centering constants can be chosen as  $A_n = n \int_{|x| \leq B_n} x dF(x)$ .

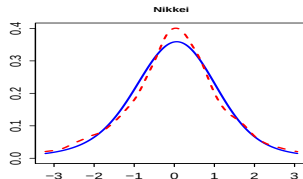
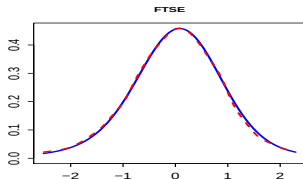
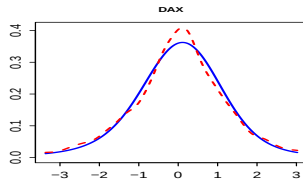
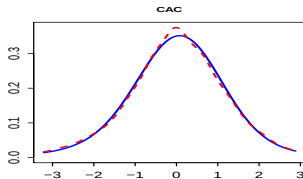


## Stable densities fitted by ML on daily indices returns

Table – Daily returns from January 1991 to August 2011. Estimated standard deviations in parentheses

| Index  | $\hat{\alpha}$ | $\hat{\beta}$ | $\hat{\sigma}$ | $\hat{\mu}$ |
|--------|----------------|---------------|----------------|-------------|
| CAC    | 1.72 (0.07)    | -0.19 (0.05)  | 0.81 (0.03)    | 0.07 (0.02) |
| DAX    | 1.64 (0.07)    | -0.17 (0.05)  | 0.79 (0.04)    | 0.09 (0.02) |
| FTSE   | 1.70 (0.06)    | -0.19 (0.04)  | 0.62 (0.02)    | 0.07 (0.01) |
| Nikkei | 1.65 (0.05)    | -0.14 (0.03)  | 0.79 (0.03)    | 0.05 (0.02) |
| NSE    | 1.60 (0.09)    | -0.21 (0.07)  | 0.87 (0.05)    | 0.17 (0.04) |
| SMI    | 1.66 (0.06)    | -0.22 (0.05)  | 0.64 (0.02)    | 0.09 (0.02) |
| SP500  | 1.62 (0.05)    | -0.10 (0.03)  | 0.50 (0.01)    | 0.05 (0.01) |
| SPTSX  | 1.55 (0.11)    | -0.25 (0.05)  | 0.60 (0.03)    | 0.11 (0.02) |
| SSE    | 1.54 (0.06)    | -0.12 (0.07)  | 0.83 (0.03)    | 0.09 (0.04) |

Comparison : estimated stable densities (full lines) vs nonparametric estimates (dotted lines)



## Temporal dependencies

Empirical autocovariance of order  $k$  : defined for  $0 \leq k < n$  by

$$\hat{\gamma}(k) = \frac{1}{n} \sum_{t=1}^{n-k} (r_t - \bar{r})(r_{t+k} - \bar{r}) := \hat{\gamma}(-k).$$

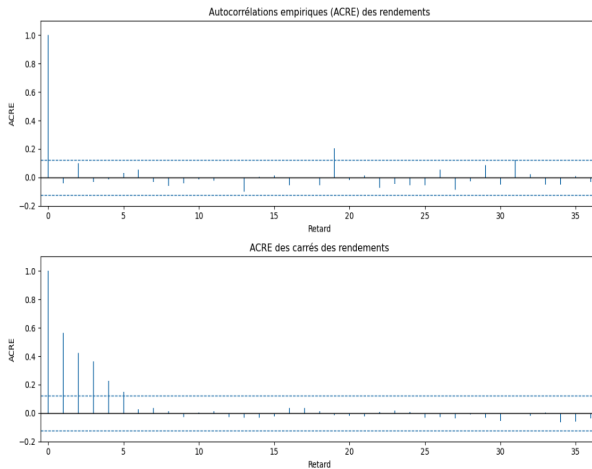
Similarly, the empirical autocorrelations of order  $k$  are defined by

$$\hat{\rho}(k) = \hat{\gamma}(k) / \hat{\gamma}(0), \quad |k| < n.$$

**Returns** : generally no (or small) autocorrelation. But the squares are highly correlated.

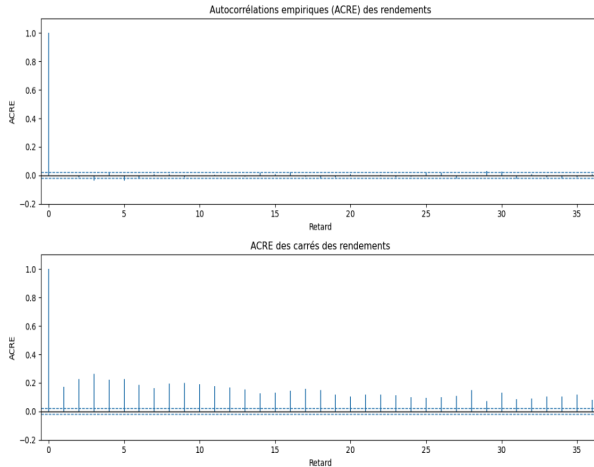
# Autocorrelations of the CAC returns and squared returns

Empirical autocorrelations for the CAC index returns  
(25/09/2024 - 25/09/2025)



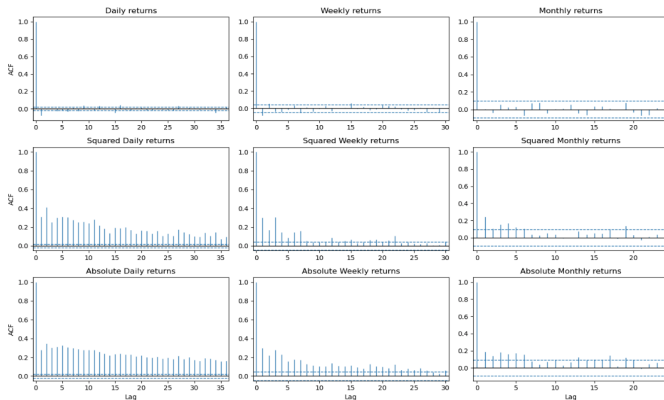
# Autocorrelations of the CAC returns and squared returns

Empirical autocorrelations for the CAC index returns  
(02/01/1990 - 25/09/2025)



# Autocorrelations of the S&P500 index returns, squared returns and absolute returns

Autocorrelations of the S&P500 index returns, squared returns and absolute returns  
02/01/1990 to 22/08/2025



## Temporal cross dependencies

For assets  $i$  and  $j$ , defined the cross empirical autocovariances of order  $k$ , for  $0 \leq k < n$  :

$$\hat{\gamma}_{ij}(k) = \frac{1}{n} \sum_{t=1}^{n-k} (r_{i,t} - \bar{r}_i)(r_{j,t+k} - \bar{r}_j),$$

and the cross empirical autocorrelations :

$$\hat{\rho}_{ij}(k) = \frac{\hat{\gamma}_{ij}(k)}{\sqrt{\hat{\gamma}_{ii}(0)\hat{\gamma}_{jj}(0)}}.$$